

Forensic-Aware Continual Adaptation with Fisher-Guided LoRA Surgery for Image Forgery Localization

Chenqi Kong¹, Song Xia^{2*}, Anwei Luo³, Harry Cheng¹

¹National University of Singapore

²Nanyang Technological University

³Jiangxi University of Finance and Economics

*Corresponding author

cqkong@nus.edu.sg, xias0002@e.ntu.edu.sg, luowanwei@jxufe.edu.cn, xaCheng1996@gmail.com

Abstract

The rapid evolution of image manipulation techniques has raised pressing public security concerns. Existing countermeasures mitigate these threats by accurately localizing manipulated regions in images. However, they are often unable to adapt dynamically to newly emerging forgeries. In real-world forensic scenarios, data typically arrive sequentially, yet existing image forgery localization (IFL) methods largely overlook the need for frequent model updates in practical workflows. To address this gap, we propose the first continual learning framework for the IFL task. We first establish a comprehensive benchmark with eight state-of-the-art baseline methods under two data evolution protocols: a cross-dataset setting and a cross-content setting. Our results show that existing methods suffer significant performance degradation under both settings. These observations reveal two key challenges in continual IFL: (1) how to adaptively capture inherent forensic traces from incoming data across diverse unseen domains; and (2) how to mitigate catastrophic forgetting when substantial domain gaps exist between current and previous tasks. To tackle these challenges, we propose **FO**rensic-aware **C**ontinual **A**daptation with **L**oRA surgery (**FOCAL**). Specifically, we design a forensic trace mining module equipped with a **S**patial **M**ixture-of-**F**orensic-**E**xperts (**SMoFE**) mechanism to dynamically extract robust forensic features. We further introduce a Fisher-weighted LoRA gradient surgery strategy, which performs importance-aware gradient modulation on LoRA parameters to preserve previously learned knowledge. Extensive experiments demonstrate that FOCAL achieves state-of-the-art performance in both pixel-level and image-level continual forgery detection.

1 Introduction

Image manipulation has long been a critical while long-standing problem, which poses grand security concerns to

a wide variety of areas, such as politics and frauds. To counter these threats, substantial efforts have been made to develop image forgery localization (IFL) techniques to capture various manipulation artifacts, including lens distortions [Mayer and Stamm, 2018; Yerushalmy and Hel-Or, 2011], Color Filter Array (CFA) artifacts [Kong *et al.*, 2025; Cao and Kot, 2009], noise patterns [Lyu *et al.*, 2014; Kobayashi *et al.*, 2010], compression artifacts [Fan and De Queiroz, 2003; Chen and Hsu, 2011], among others. Beyond these hand-crafted features, later learning-based methods built upon advanced architectures (e.g., SAM [Kirillov *et al.*, 2023] and Swin [Liu *et al.*, 2021]) have achieved higher localization accuracy. Recent studies have further focused on improving model generalization to unseen datasets and robustness against common image distortions.

Despite this progress, existing IFL models still exhibit limited generalizability when encountering newly emerging forgeries. This limitation has become more severe with the advent of foundation models, such as GPT-image 2.0, which can generate highly realistic content and blur the boundary between authentic and manipulated images. These limitations mainly arise from two challenges. First, newly emerging forgeries are increasingly sophisticated and often leave fewer detectable traces for existing models. Second, new manipulation techniques can be applied to substantially different image domains, such as document and scientific images, resulting in large domain gaps. As foundation models rapidly evolve, IFL defenses often lag behind emerging attacks. Therefore, IFL models must be frequently updated to adapt to new forgeries.

However, state-of-the-art IFL models, particularly SAM-based methods [Kwon *et al.*, 2025], typically contain a large number of parameters and are trained on datasets with millions of images. Retraining the entire model on the full dataset whenever new forgeries emerge is computationally expensive and impractical for real-world deployment. This motivates the need for a continual IFL paradigm that can progressively adapt IFL models to newly emerging forgeries while retaining previously learned knowledge. Different from prior studies, this work reformulates the conventional IFL task as a more practical continual learning problem.

To this end, we first conduct an extensive benchmark of state-of-the-art continual learning and IFL methods under two evaluation protocols. **Protocol 1** consists of four image manipulation datasets with diverse forgery types, namely Clas-

sis [Dong *et al.*, 2013; IFC, 2013; Zampoglou *et al.*, 2015; Novozamsky *et al.*, 2020], Defacto [Mahfoudi *et al.*, 2019], FantasticReality [Kniaz *et al.*, 2019], and TampCOCO [Kwon *et al.*, 2022], sorted according to their release dates. This protocol simulates practical scenarios in which datasets arrive sequentially. **Protocol 2** covers three types of image content: natural images [Dong *et al.*, 2013; IFC, 2013; Zampoglou *et al.*, 2015; Novozamsky *et al.*, 2020], document images [Qu *et al.*, 2023; Yu *et al.*, 2023], and scientific images [Cardenuto and Rocha, 2022; Sabir *et al.*, 2021; Lin *et al.*, 2024]. This protocol reflects more complex real-world scenarios and introduces more pronounced domain gaps. We benchmark SOTA continual learning and IFL techniques under both protocols and found that these methods suffer significant performance degradation in the continual learning paradigm.

Based on these observations, we identify two critical challenges in continual IFL: (1) how to adaptively capture inherent forensic traces from incoming data across diverse unseen domains; and (2) how to mitigate catastrophic forgetting when substantial domain gaps exist between current and previous tasks. To address these challenges, we propose a forensic trace mining module equipped with a Spatial Mixture-of-Forensic-Experts (SMoFE) mechanism, which dynamically extracts robust forensic features for diverse unseen domains. In addition, we introduce Fisher-weighted LoRA gradient surgery, which performs importance-aware gradient modulation on LoRA parameters to mitigate catastrophic forgetting.

The main contributions of this work are:

- We are the first to benchmark continual IFL under cross-dataset and cross-content protocols. The results reveal that existing IFL models suffer severe performance degradation in continual learning scenarios.
- We design a forensic trace mining module that adaptively integrates local anomalies from different forensic maps through a MoE mechanism, enabling forensic knowledge to be efficiently adapted to the SAM backbone.
- We propose Fisher-weighted LoRA gradient surgery for continual adaptation, which adaptively modulates the gradients of different LoRA parameters in the SAM encoder and effectively mitigates catastrophic forgetting on previous tasks.

2 RELATED WORKS

With the rapid development of AIGC techniques, IFL systems inevitably encounter test samples from unseen domains, making continual adaptation to newly collected data essential. Continual learning methods can be broadly categorized into three groups: (1) regularization-based methods [Wang *et al.*, 2026; Kirkpatrick *et al.*, 2017], which introduce additional regularization terms to consolidate previously learned knowledge while adapting models to new tasks; (2) replay-based methods [Zhanwang *et al.*, 2026; Kapoor *et al.*, 2026], which selectively store samples from previous tasks in a memory bank and replay them during training on the current task; and (3) parameter-isolation methods [Zhang *et al.*, 2026; Ye *et al.*, 2026], which typically

rely on a task oracle to design task-specific architectures or allocate task-specific parameters. Recent efforts in continual forensic learning have covered a wide range of applications, including face anti-spoofing [Cai *et al.*, 2023; Rostami *et al.*, 2021], AIGC detection [Tang *et al.*, 2025; Marra *et al.*, 2019], and face forgery detection [Cheng *et al.*, 2025; Zhang *et al.*, 2025]. Although these methods address continual binary classification and alleviate catastrophic forgetting to some extent, continual image forgery localization remains largely unexplored. Unlike binary classification, IFL is a dense prediction task that requires pixel-level localization of manipulated regions, making continual adaptation substantially more challenging. Moreover, existing continual forensic learning methods often fail to explicitly capture inherent forensic traces from forgery data, leading to suboptimal localization performance. In contrast, this work introduces a Spatial Mixture-of-Forensic-Experts module and Fisher-weighted LoRA gradient surgery to extract robust forensic features while mitigating catastrophic forgetting.

3 Methodology

3.1 Framework Overview

Figure 1 illustrates the overall framework of the proposed methodology, which consists of an image encoder $E_{\theta,\phi}$, a forensic prompt learner, and a mask decoder D_ψ , where we freeze the image encoder as the pretrained weights while only updating LoRA modules during training. The designed Forensic Trace Mining module consists of a Spatial Mixture-of-Forensic-Expert (SMoFE) module and a Forensic Evidence-Guided Dense Prompting (FEGDP) module. SMoFE adaptively combines local anomalies from different forensic maps via MoE mechanisms, and FEGDP aims to mine the forensic knowledge which can be fast adapted to the SAM backbone. We design a Fisher-Weighted LoRA Gradient Surgery for Continual Adaptation, which adaptively modulates the gradients for different LoRA parameters in the image encoder, to counter the catastrophic forgetting issues.

3.2 Forensic Trace Mining with Spatial Mixture-of-Forensic-Expert (SMoFE)

Spatial Routing of Forensic Trace Expert. We build our model upon the Segment Anything Model (SAM) and adapt it to image forgery localization by learning forensic prompts. Given an input image $\mathbf{x} \in \mathbb{R}^{3 \times H \times W}$, SAM first extracts visual embeddings

$$\mathbf{z}^{\text{img}} = E_{\theta,\phi}(\mathbf{x}) \in \mathbb{R}^{C \times h \times w}, \quad (1)$$

where $E_{\theta,\phi}$ denotes the SAM image encoder equipped with LoRA adapters. The pretrained SAM parameters θ are frozen, while the low-rank adapters ϕ are trainable.

There are two challenges in the continual image forgery localization paradigm: (1) How to expose manipulation artifacts that are not explicitly modeled by natural-image segmentation backbones? (2) How to adaptively extract the most appropriate forensic features of newly coming data from unseen domains?

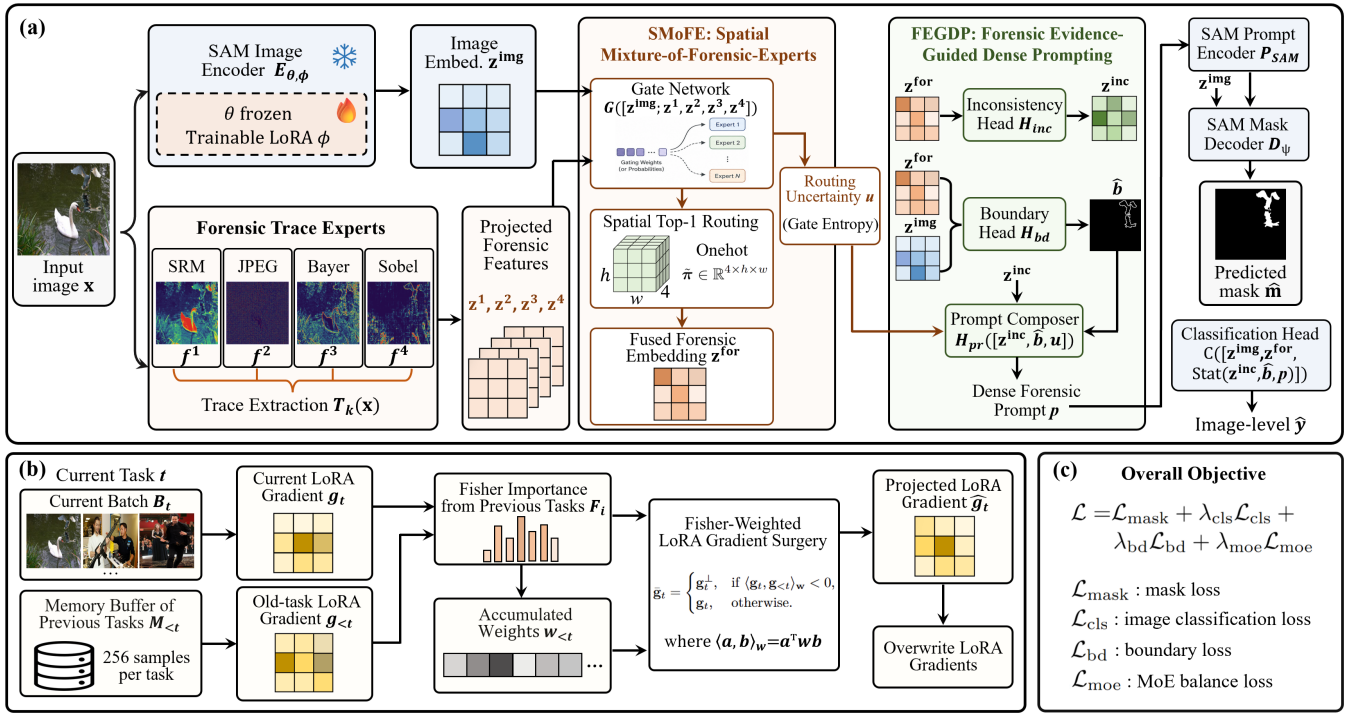


Figure 1: Overall framework of the proposed Forensic-aware Continual Adaptation with LoRA surgery (FOCAL).

In this work, we introduce a forensic trace branch with multiple artifact experts. Specifically, we use four complementary trace extractors, which have been widely proven effective in previous IFL works:

$$\mathcal{E} = \{\text{Sobel}, \text{SRM}, \text{JPEG}, \text{Bayer}\}, \quad (2)$$

These forensic experts respectively capture edge discontinuities, high-frequency residuals, compression blockiness, and CFA/Bayer-pattern inconsistencies. For each expert $k \in \mathcal{E}$, a handcrafted forensic trace prior is first extracted and then projected to the SAM embedding space:

$$\mathbf{z}^k = P_k(T_k(\mathbf{x})) \in \mathbb{R}^{C \times h \times w}, \quad (3)$$

where $T_k(\cdot)$ is the k -th forensic trace extractor and $P_k(\cdot)$ is a learnable convolutional projector.

Different manipulation types tend to expose forensic artifacts in different local regions, to complementarily combine these abundant forensic traces, we design a novel spatial Top-1 mixture-of-experts routing mechanism. Unlike previous MoE works that usually select the optimal expert map along the channel dimension, we perform element-wise MoE in the spatial feature plane. Due to the geometric alignment among the four forensic trace maps, our designed SMoFE can adaptively combine the local anomaly of these traces.

We design a gate network to predict expert probabilities conditioned on both semantic visual features and forensic trace features:

$$\boldsymbol{\pi} = \text{softmax}(G([\mathbf{z}^{\text{img}}; \mathbf{z}^1, \dots, \mathbf{z}^K])), \quad \boldsymbol{\pi} \in \mathbb{R}^{K \times h \times w}. \quad (4)$$

The SAM image embedding $\mathbf{z}^{\text{img}}$ is incorporated into the MoE router as contextual guidance rather than as an additional expert. While the forensic experts capture low-level

manipulation traces such as edges, noise residuals, compression artifacts, and CFA inconsistencies, the image embedding provides semantic and structural context from the visual backbone. By concatenating image and expert embeddings, the router can make spatially adaptive expert selections conditioned on both forensic evidence and image content. This allows the model to suppress unreliable trace responses in benign regions and emphasize the most relevant forensic cue around manipulated areas. We use a straight-through Top-1 routing strategy:

$$\tilde{\boldsymbol{\pi}} = \text{onehot}\left(\arg \max_k \pi_k\right). \quad (5)$$

The fused forensic trace embedding is then computed as:

$$\mathbf{z}^{\text{for}} = \sum_{k=1}^K \tilde{\pi}_k \odot \mathbf{z}^k. \quad (6)$$

In this vein, the fused forensic map combines more diverse local artifacts as a whole. Compared to simply concatenating the four forensic maps, the adopted SMoFE scheme can dynamically select the most appropriate features for newly coming data from unseen domains.

Forensic Evidence-Guided Dense Prompting. The vanilla SAM takes image embedding and meaningful dense/sparse prompts (e.g., mask and bounding box) as inputs. Previous SAM-based IFL methods use low-level forensic cues as prompt, resulting in difficulties for SAM decoder to understand the prompt features, leading to inferior performance.

To minimize this gap, we design an inconsistency-modeling head $H_{\text{inc}}()$ and a boundary-capturing head $H_{\text{bd}}()$ which bridge the low-level forensic cues to meaningful

forgeries boundaries in a coarse-to-fine manner. We first predict an inconsistency evidence map from the fused forensic embedding:

$$\mathbf{z}^{\text{inc}} = H_{\text{inc}}(\mathbf{z}^{\text{for}}), \quad (7)$$

where H_{inc} measures local feature inconsistency through multi-scale neighborhood statistics. In particular, for each spatial scale $s \in \{3, 5, 7\}$, we compare the local feature with its neighborhood mean and aggregate cosine dissimilarity and local variance cues. This encourages the model to highlight regions whose forensic traces are locally inconsistent with their surroundings.

We further estimate boundary evidence by comparing forensic and visual gradients:

$$\hat{\mathbf{b}} = H_{\text{bd}}(\mathbf{z}^{\text{for}}, \mathbf{z}^{\text{img}}, \mathbf{z}^{\text{inc}}), \quad (8)$$

where H_{bd} takes the forensic features $\mathbf{z}^{\text{for}}$, the image embeddings $\mathbf{z}^{\text{img}}$, and the inconsistency map $\mathbf{z}^{\text{inc}}$ as inputs. This boundary branch is designed to capture transition artifacts around manipulated regions.

The final forensic dense prompt is composed from four evidence cues: inconsistency probability, boundary probability, a locally expanded boundary band, and routing uncertainty. The routing uncertainty is measured by the normalized gate entropy:

$$\mathbf{u} = -\frac{1}{\log K} \sum_{k=1}^K \pi_k \log(\pi_k + \epsilon). \quad (9)$$

The normalized gate entropy $\mathbf{u}$ measures the uncertainty of the forensic MoE router at each spatial location. Low entropy indicates that one trace expert dominates, while high entropy suggests ambiguous or mixed forensic evidence across Sobel, SRM, JPEG, and Bayer cues. We use this entropy map as an additional input to the prompt composer, together with inconsistency and boundary evidence, so that the generated dense prompt can adapt to regions where expert selection is uncertain. The prompt composer is a simple 3-layer convolution network, which predicts a dense prompt logit map:

$$\mathbf{p} = H_{\text{pr}}([\mathbf{z}^{\text{inc}}, \hat{\mathbf{b}}, \mathbf{u}]), \quad (10)$$

where $[\cdot]$ denotes the concatenation operation along the channel dimension. This forensic prompt is then encoded by SAM's prompt encoder and injected into the SAM mask decoder, and the predicted forgery mask is computed by:

$$\hat{\mathbf{m}} = D_{\psi}(\mathbf{z}^{\text{img}}, P_{\text{SAM}}(\mathbf{p})), \quad (11)$$

where P_{SAM} is the SAM prompt encoder and D_{ψ} is the SAM mask decoder. Thus, instead of relying on manual point or box prompts, our model learns more meaningful prompts from forensic evidences.

3.3 Fisher-Weighted LoRA Gradient Surgery for Continual Adaptation

In the continual forgery localization paradigm, the model observes a sequence of datasets or tasks $\{\mathcal{D}_1, \mathcal{D}_2, \dots, \mathcal{D}_T\}$. After training on task t , the model should preserve its performance on previous tasks while adapting to the current one.

How to solve the catastrophic forgetting issues is another challenge for this task.

Instead of applying replay or regularization to the entire network, we constrain only the LoRA subspace of the SAM image encoder. This keeps the protection mechanism focused on the parameters responsible for domain adaptation, while leaving the rest of the trainable components optimized by the current task objective. This choice is motivated by the fact that LoRA parameters provide the main task-adaptive capacity of the frozen SAM encoder.

Let ϕ denote the collection of trainable LoRA parameters. For each completed task $i < t$, we store a small memory buffer $\mathcal{M}_i \subset \mathcal{D}_i$ (256 samples for each task). At the beginning of task t , the memory of all previous tasks is:

$$\mathcal{M}_{<t} = \bigcup_{i=1}^{t-1} \mathcal{M}_i. \quad (12)$$

During training on a current mini-batch $\mathcal{B}_t$, we first compute the ordinary task loss:

$$\mathcal{L}_t = \mathcal{L}(\mathcal{B}_t; \Theta), \quad (13)$$

where Θ denotes all trainable parameters. Back-propagation gives gradients for the entire model, but we only extract the LoRA gradients:

$$\mathbf{g}_t = \nabla_{\phi} \mathcal{L}_t. \quad (14)$$

We then sample a reference mini-batch $\mathcal{B}_{<t}$ (default: 16) from $\mathcal{M}_{<t}$ and compute the old-task LoRA gradient:

$$\mathbf{g}_{<t} = \nabla_{\phi} \mathcal{L}(\mathcal{B}_{<t}; \Theta). \quad (15)$$

To emphasize LoRA dimensions that are important for previous tasks, we estimate a diagonal Fisher importance vector after each task, which indicates the importance of each LoRA parameter. For task i , after loading the checkpoint of that task, we compute

$$\mathbf{F}_i = \mathbb{E}_{\mathcal{B} \sim \mathcal{M}_i} [(\nabla_{\phi} \mathcal{L}(\mathcal{B}; \Theta))^2], \quad (16)$$

where the square is taken element-wise. The Fisher vector is normalized by its mean and clipped:

$$\tilde{\mathbf{F}}_i = \text{Clip} \left(\frac{\mathbf{F}_i}{\text{mean}(\mathbf{F}_i) + \epsilon}, F_{\min}, F_{\max} \right). \quad (17)$$

For the current task t , the accumulated importance vector is the average over previous tasks:

$$\mathbf{w}_{<t} = \frac{1}{t-1} \sum_{i=1}^{t-1} \tilde{\mathbf{F}}_i. \quad (18)$$

We further propose a Fisher-weighted projection geometry. Specifically, after finishing each task, we estimate a diagonal Fisher importance vector over LoRA parameters using the squared gradients on that task's memory samples. The resulting importance vector assigns larger weights to LoRA dimensions to which the old-task objective is more sensitive. This design makes conflict detection sensitive to old-task critical directions. Conflicts occurring on Fisher-important LoRA dimensions are emphasized, whereas disagreements on less important dimensions have a weaker effect. As a result, the

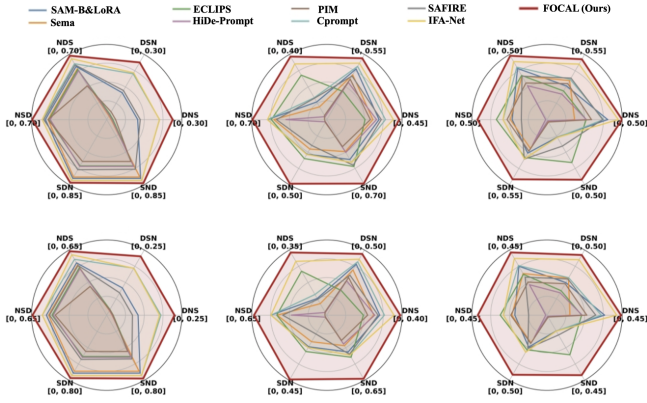


Figure 2: Experimental results (Average F1) under Protocol 2.

method protects parameters that are more relevant for previous forensic domains while preserving sufficient plasticity for learning the current domain.

Gradient conflict is detected by the sign of the weighted inner product:

$$s = \langle \mathbf{g}_t, \mathbf{g}_{<t} \rangle_{\mathbf{w}} = \sum_j w^j \mathbf{g}_t^j \mathbf{g}_{<t}^j. \quad (19)$$

A negative inner product ($s < 0$) between them indicates gradient conflict: updating along $\mathbf{g}_t$ may increase the loss on previous tasks. Previous method resolves this by removing the component of $\mathbf{g}_t$ that conflicts with $\mathbf{g}_{<t}$. However, not all LoRA parameters are equally important for retaining old-task knowledge. Treating every LoRA dimension uniformly may either under-protect sensitive parameters or over-constrain less relevant ones.

If $s \geq 0$, the current LoRA update is considered compatible with previous tasks and is left unchanged. If $s < 0$, we remove from the current gradient the component that conflicts with the old-task gradient:

$$\mathbf{g}_t^\perp = \mathbf{g}_t - \frac{\langle \mathbf{g}_t, \mathbf{g}_{<t} \rangle_{\mathbf{w}}}{\|\mathbf{g}_{<t}\|_{\mathbf{w}}^2 + \epsilon} \mathbf{g}_{<t}. \quad (20)$$

This design makes conflict detection sensitive to old-task critical directions. Conflicts occurring on Fisher-important LoRA dimensions are emphasized, whereas disagreements on less important dimensions have a weaker effect. As a result, the method protects parameters that are more relevant for previous forensic domains while preserving sufficient plasticity for learning the current domain. Thus, the final LoRA gradient is modulated as:

$$\bar{\mathbf{g}}_t = \begin{cases} \mathbf{g}_t^\perp, & \text{if } \langle \mathbf{g}_t, \mathbf{g}_{<t} \rangle_{\mathbf{w}} < 0, \\ \mathbf{g}_t, & \text{otherwise.} \end{cases} \quad (21)$$

After projection, we overwrite only the gradients of LoRA parameters:

$$\nabla_{\phi} \mathcal{L}_t \leftarrow \bar{\mathbf{g}}_t. \quad (22)$$

All other trainable parameters, including the SAM mask decoder, the prompt encoder mask-downscaling module, the forensic trace experts, the prompt composer, and the classification head, retain their ordinary current-task gradients. The

Table 1: Image numbers of the training and testing sets for the four datasets under Protocol 1.

Dataset	Sum	Train	Test
Classic	25,296	19,735	6,264
DEFACTO	27,106	21,106	6,000
FantasticReality	36,000	26,677	9,323
TampCOCO	48,445	48,445	12,000

Table 2: Image numbers of the training and testing sets for the three datasets under Protocol 2.

Dataset	Sum	Train	Test
Natural	16,946	12,582	4,364
Scientific	8,131	5,671	2,460
Document	16,432	12,750	3,682

optimizer then performs a standard update over all trainable parameters:

$$\Theta \leftarrow \Theta - \eta \bar{\nabla}_{\Theta}, \quad (23)$$

where $\bar{\nabla}_{\Theta}$ equals the projected gradient for LoRA parameters and the original current-task gradient for all other parameters.

Overall, the proposed Fisher-weighted LoRA gradient surgery is particularly suited to continual forensic segmentation, where each task may correspond to a different manipulation distribution. These domains often rely on different forensic cues, including boundary inconsistencies, texture residuals, compression artifacts, and semantic-context mismatches. Instead of constraining the entire model, our method operates only in the LoRA subspace, where task-specific adaptation is concentrated.

3.4 Overall Objectives

The overall training objective $\mathcal{L}$ consists of the following four parts: the mask prediction loss $\mathcal{L}_{\text{mask}}$, the image classification loss $\mathcal{L}_{\text{cls}}$, the boundary prediction loss $\mathcal{L}_{\text{bd}}$, and the MoE loss $\mathcal{L}_{\text{moe}}$:

$$\mathcal{L} = \mathcal{L}_{\text{mask}} + \lambda_{\text{cls}} \mathcal{L}_{\text{cls}} + \lambda_{\text{bd}} \mathcal{L}_{\text{bd}} + \lambda_{\text{moe}} \mathcal{L}_{\text{moe}}, \quad (24)$$

where $\mathcal{L}_{\text{mask}}$ and $\mathcal{L}_{\text{cls}}$ compute the binary cross-entropy losses between the predicted mask $\hat{\mathbf{m}}$ and class $\hat{\mathbf{y}}$ and the ground-truth manipulation mask $\mathbf{m}$ and image-level label $y \in \{0, 1\}$, respectively. We aggregate global visual and forensic features together with evidence-map statistics to predict the image-level manipulation score:

$$\hat{y} = C \left([\mathbf{z}^{\text{img}}, \mathbf{z}^{\text{for}}, \text{Stat}(\mathbf{z}^{\text{inc}}, \hat{\mathbf{b}}, \mathbf{p})] \right), \quad (25)$$

where $\text{Stat}(\cdot)$ includes the mean and top- k mean of each evidence map, and C represents two convolutional layers. The boundary loss supervises the predicted boundary $\hat{\mathbf{b}}$ with the boundary label $\mathbf{b}$:

$$\mathcal{L}_{\text{bd}} = \text{Focal}(\hat{\mathbf{b}}, \mathbf{b}) + \text{Dice}(\hat{\mathbf{b}}, \mathbf{b}) + \|\hat{\mathbf{b}} - \partial \hat{\mathbf{m}}\|_1, \quad (26)$$

where the third component makes the boundary evidence consistent with the final SAM prediction, $\partial \hat{\mathbf{m}}$ calculates the boundary of the predicted forgery mask. Finally, the expert balance loss prevents the Top-1 router from collapsing to a single forensic expert:

$$\mathcal{L}_{\text{moe}} = \left\| \frac{1}{hw} \sum_{i,j} \pi_{:,i,j} - \frac{1}{K} \mathbf{1} \right\|_2^2. \quad (27)$$

4 Experiments

4.1 Experimental Settings

Protocol 1 simulates the practical scenario that model adapts as new dataset emerges. As such, model under Protocol 1 evolves in the following sequential tasks: **Classic**→**DEFACTO**→**FantasticReality**→**TampCOCO**, sorted by released date, where Classic incorporates four traditional manipulation datasets: CASIAv2, IFC, WildWeb, IMD2020. These datasets include diverse manipulation types (e.g., splicing, copy-move, and inpainting) with various forgery region areas. We carefully breakdown the training and testing sets to avoid information leakage. The image numbers of the training and testing sets are shown in Table 1.

Protocol 2 is tailored to assess IFL models’ capability when facing novel forgery content, which encompasses three tasks: Natural, Scientific, and Document data. We collect natural image data from the following four datasets: CASIAv2, IFC, WildWeb, IMD2020. The scientific data consists of three datasets: RSIID, Biofors, SciSp. And we combine STFD and part of DocTammer as the document data. Similarly, we carefully breakdown the training and testing sets to avoid information leakage. The image numbers of the training and testing sets are shown in Table 2. We evaluate all six possible task orders in a continual manner under Protocol 2.

Implementation Details

Our model is implemented in PyTorch [Paszke *et al.*, 2019] and trained on one NVIDIA RTX PRO 6000 GPU. We adopt the SAM ViT-B backbone initialized from the pretrained SAM checkpoint, and insert LoRA modules with rank 32 into the image encoder. The input image size is 512×512 and the mask resolution is 128×128 . We use the Adam optimizer [Kingma and Ba, 2015] with $\beta_1=0.9$ and $\beta_2=0.999$ to train the model with batch size 64. The learning rate and weight decay are $1e-4$ and $1e-5$, respectively. The model is trained for 10 epochs for each continual-learning task and validated after every epoch on all seen validation sets.

4.2 Experimental Results under Protocol 1

The results in Table 3 demonstrate the superiority of the proposed FOCAL framework in continual forgery detection at both pixel and image levels. Compared with existing continual learning methods and IFL baselines, FOCAL consistently achieves the best average performance across the sequential tasks, especially at the final stage T4, where all datasets have been incrementally introduced. At the pixel level, FOCAL obtains the highest average F1 and IoU, indicating its stronger ability to preserve localization knowledge while adapting to newly arriving forgery domains. At the image level, FOCAL also achieves the best average AUC, showing that the learned forensic representations are effective not only for dense localization but also for global forgery discrimination. The performance gains can be attributed to the complementary designs of SMoFE and Fisher-weighted LoRA gradient surgery. Specifically, SMoFE adaptively mines diverse

forensic traces from heterogeneous manipulation domains, enabling the model to capture robust low-level evidence for new tasks. Meanwhile, Fisher-weighted LoRA Adaptive Gradient Surgery (FLAG) modulates the update of task-relevant parameters according to their importance, thereby reducing catastrophic forgetting of previously learned knowledge.

4.3 Experimental Results under Protocol 2

As shown in Fig. 2, the proposed FOCAL achieves the largest radar coverage across all six task orders under Protocol 2, indicating its strong robustness when facing novel forgery contents from natural, scientific, and document domains. Compared with existing IFL methods, FOCAL consistently stays close to the outer boundary on different order permutations, while the competing methods show obvious fluctuations across task orders. This suggests that FOCAL is less sensitive to the order of incoming domains and can better preserve previously learned forgery knowledge while adapting to new content types. In contrast, methods such as SAM-B&LoRA, HiDe-Prompt, and PIM suffer from unstable performance under certain orders, revealing their limited ability to handle large domain shifts. IFA-Net and SAFIRE provide relatively competitive results, but they still fall behind FOCAL in terms of overall stability and coverage. These results demonstrate the effectiveness of FOCAL in content-level incremental forgery localization, especially under challenging cross-domain continual learning scenarios.

4.4 Ablation Experiments

Effectiveness of the Designed Components

FOCAL consists of Spatial Mixture-of-Forensic-Expert (SMoFE) and Fisher-Weighted LoRA Gradient Surgery (FLAG) modules, where SMoFE aims to adaptively extract robust forensic features, and FLAG aims to preserve the previously learned knowledge. We present the evaluation results under Protocol 1 in Table 4. The full FOCAL framework achieves consistently strong average performance across continual tasks, especially on T2 and T4. Removing SMoFE causes the most significant degradation, indicating its importance for task-adaptive forgery representation learning and continual adaptation. Removing FLAG also leads to lower overall performance, showing that it provides complementary guidance for improving robustness. These results verify that both modules contribute to the final performance, with SMoFE playing a particularly critical role.

Effectiveness of the Forensic Experts

The ablation results in Table 5 validate the effectiveness of the proposed Spatial Mixture-of-Forensic-Experts (SMoFE) module in continual image forgery localization. The full FOCAL model consistently achieves the best overall performance across the continual learning sequence, especially at the final stage T4, where it outperforms all expert-removal variants in terms of average F1 and IoU. This demonstrates that SMoFE effectively mitigates performance degradation during continual adaptation by dynamically integrating complementary forensic cues. Removing any expert weakens the model’s overall ability to mine robust forensic traces. These results indicate that SMoFE can adaptively capture diverse low-level forensic evidence, including demosaicing artifacts,

Table 3: Continual forgery detection performance on sequential tasks (T1-T4) under Protocol 1. Pixel-level F1 and IoU, and image-level AUC are reported.

Method	Task	Classic			DEFACTO			FantasticReality			TampCOCO			Average		
		F1	IoU	AUC	F1	IoU	AUC	F1	IoU	AUC	F1	IoU	AUC	F1	IoU	AUC
HiDe-Prompt (NeurIPS'23)	T1	0.4955	0.4196	0.8472	-	-	-	-	-	-	-	-	-	0.4955	0.4196	0.8472
	T2	0.4758	0.4085	0.7577	0.1378	0.1051	0.6052	-	-	-	-	-	-	0.3068	0.2568	0.6815
	T3	0.4661	0.4001	0.7337	0.1342	0.1023	0.6044	0.1723	0.1148	0.7931	-	-	-	0.2575	0.2057	0.7104
	T4	0.4614	0.3959	0.7269	0.1317	0.1007	0.6030	0.2169	0.1539	0.8122	0.4156	0.3630	0.7751	0.3064	0.2534	0.7293
Eclipse (CVPR'24)	T1	0.4918	0.4363	0.8618	-	-	-	-	-	-	-	-	-	0.4918	0.4363	0.8618
	T2	0.4641	0.4166	0.7789	0.0935	0.0723	0.6425	-	-	-	-	-	-	0.2788	0.2444	0.7107
	T3	0.3688	0.3232	0.7322	0.0983	0.0746	0.6025	0.1858	0.1250	0.8840	-	-	-	0.2176	0.1743	0.7396
	T4	0.3647	0.3196	0.7249	0.0957	0.0725	0.6027	0.2116	0.1465	0.8932	0.3187	0.2702	0.7747	0.2477	0.2022	0.7489
Cprompt (CVPR'24)	T1	0.5821	0.5363	0.9618	-	-	-	-	-	-	-	-	-	0.5821	0.5363	0.9618
	T2	0.0914	0.0746	0.5272	0.2767	0.2431	0.7998	-	-	-	-	-	-	0.1841	0.1588	0.6635
	T3	0.1329	0.1071	0.4168	0.0998	0.0781	0.6493	0.6593	0.5714	0.9998	-	-	-	0.2973	0.2522	0.6886
	T4	0.0824	0.0680	0.5124	0.1156	0.0978	0.6678	0.6188	0.5574	0.9835	0.8527	0.8253	0.9798	0.4174	0.3871	0.7859
SEMA (CVPR'25)	T1	0.4934	0.4565	0.9317	-	-	-	-	-	-	-	-	-	0.4934	0.4565	0.9317
	T2	0.0614	0.0490	0.5145	0.2343	0.2044	0.7657	-	-	-	-	-	-	0.1479	0.1267	0.6401
	T3	0.0943	0.0721	0.6185	0.0813	0.0599	0.5821	0.4556	0.3617	0.9918	-	-	-	0.2104	0.1646	0.7308
	T4	0.0412	0.0301	0.6179	0.1067	0.0873	0.6508	0.2912	0.2384	0.9577	0.6747	0.6296	0.9288	0.2784	0.2464	0.7888
PIM (TPAMI'25)	T1	0.3836	0.3180	0.8420	-	-	-	-	-	-	-	-	-	0.3836	0.3180	0.8420
	T2	0.2941	0.2190	0.7906	0.1691	0.1262	0.6184	-	-	-	-	-	-	0.2316	0.1726	0.7045
	T3	0.2559	0.1756	0.3668	0.1235	0.0814	0.4974	0.4949	0.3694	0.9966	-	-	-	0.2914	0.2088	0.6203
	T4	0.1363	0.0913	0.5902	0.0618	0.0429	0.5340	0.3702	0.2755	0.9237	0.6249	0.5491	0.8827	0.2983	0.2397	0.7327
SAFIRE (AAAI'25)	T1	0.5819	0.4979	0.8786	-	-	-	-	-	-	-	-	-	0.5819	0.4979	0.8786
	T2	0.1241	0.0794	0.6680	0.1811	0.1337	0.6664	-	-	-	-	-	-	0.1526	0.1065	0.6672
	T3	0.1865	0.1217	0.7413	0.1217	0.0821	0.5076	0.3597	0.2508	0.9674	-	-	-	0.2226	0.1515	0.7388
	T4	0.1110	0.0690	0.6568	0.0773	0.0512	0.5203	0.2424	0.1534	0.8359	0.4101	0.2955	0.8412	0.2102	0.1423	0.7136
IFA-Net (CVPR'26)	T1	0.5228	0.4587	0.8706	-	-	-	-	-	-	-	-	-	0.5228	0.4587	0.8706
	T2	0.1190	0.0820	0.5715	0.1660	0.1304	0.6901	-	-	-	-	-	-	0.1425	0.1062	0.6308
	T3	0.1249	0.0833	0.6705	0.0797	0.0539	0.5417	0.3776	0.2693	0.9782	-	-	-	0.1940	0.1355	0.7301
	T4	0.1282	0.0867	0.5681	0.0780	0.0567	0.5770	0.3816	0.2931	0.9110	0.6561	0.5775	0.9094	0.3110	0.2535	0.7414
SAM-B&LoRA (Baseline)	T1	0.4606	0.4168	0.8966	-	-	-	-	-	-	-	-	-	0.4606	0.4168	0.8966
	T2	0.0365	0.0292	0.5891	0.2116	0.1871	0.7716	-	-	-	-	-	-	0.1241	0.1081	0.6803
	T3	0.1695	0.1369	0.5238	0.0900	0.0901	0.6185	0.6989	0.6151	0.9990	-	-	-	0.3195	0.2807	0.7138
	T4	0.0841	0.0683	0.4905	0.1116	0.0947	0.6633	0.5819	0.5182	0.9884	0.8197	0.7927	0.9714	0.3993	0.3685	0.7784
FOCAL (Ours)	T1	0.6368	0.5871	0.9525	-	-	-	-	-	-	-	-	-	0.6368	0.5871	0.9525
	T2	0.4286	0.3729	0.8401	0.2921	0.2553	0.7720	-	-	-	-	-	-	0.3604	0.3141	0.8060
	T3	0.3578	0.3065	0.7232	0.1700	0.1388	0.6422	0.7075	0.6282	0.9998	-	-	-	0.4117	0.3579	0.7884
	T4	0.3787	0.3319	0.7865	0.1675	0.1417	0.6792	0.6960	0.6264	0.9905	0.8130	0.7834	0.9630	0.5138	0.4708	0.8548

Table 4: Effectiveness of the designed modules (Protocol 1). Only average results are reported.

Task	FOCAL		w/o SMoFE		w/o FLAG	
	F1	IoU	F1	IoU	F1	IoU
T1	<u>0.6801</u>	<u>0.6310</u>	0.6871	0.6364	0.6724	0.6241
T2	0.4450	0.3926	0.2911	0.2543	<u>0.4067</u>	<u>0.3579</u>
T3	<u>0.4319</u>	<u>0.3756</u>	0.3474	0.3005	0.4502	0.3922
T4	0.5722	0.5324	0.4386	0.4069	<u>0.5398</u>	<u>0.5029</u>

Table 5: Effectiveness of different forensic experts (Protocol 1). Only average results are reported.

Task	FOCAL		w/o Sobel		w/o SRM		w/o JPEG		w/o Bayer	
	F1	IoU	F1	IoU	F1	IoU	F1	IoU	F1	IoU
T1	0.6801	0.6310	0.6571	0.6096	0.6753	0.6253	0.6842	0.6336	0.6818	0.6330
T2	0.4450	0.3926	0.3917	0.3467	0.3867	0.3408	0.3812	0.3335	0.3835	0.3387
T3	0.4319	0.3756	0.4683	0.4106	0.4122	0.3594	<u>0.4416</u>	<u>0.3881</u>	0.3612	0.3107
T4	0.5722	0.5324	0.5267	0.4932	0.5143	0.4783	<u>0.5323</u>	<u>0.4975</u>	0.4827	0.4485

noise residuals, boundaries, and compression traces, thereby enhancing both forensic feature mining and continual learning performance.

5 Conclusion

In this work, we present the first continual learning framework for image forgery localization, aiming to adapt IFL models to sequentially arriving forgery data while preserving previously acquired forensic knowledge. We first establish a comprehensive benchmark under two realistic data

evolution protocols, including a cross-dataset protocol and a cross-content protocol, to systematically evaluate the continual adaptation ability of existing IFL and continual learning methods. The experimental results reveal that existing methods suffer from severe performance degradation when facing newly emerging forgery domains and substantial domain gaps. To address these challenges, we propose FOCAL, which consists of a forensic trace mining module and a Fisher-weighted LoRA gradient surgery strategy. Specifically, the SMoFE mechanism adaptively integrates complementary forensic cues from different trace experts, enabling the model to capture robust manipulation evidence from diverse unseen domains. Meanwhile, Fisher-weighted LoRA gradient surgery selectively modulates the gradients of LoRA parameters according to their importance to previous tasks, thereby mitigating catastrophic forgetting while maintaining sufficient plasticity for learning new forgery patterns. Extensive quantitative and qualitative experiments under the proposed protocols demonstrate that FOCAL consistently achieves superior performance in both pixel-level forgery localization and image-level forgery detection. These results suggest that continual image forgery localization is a practical and important research direction for real-world multimedia forensics, and we hope this work can provide a strong benchmark and baseline for future studies.

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